

Doc 2: Functions, Slopes, and Sigma

Module 0: Warm-up | Theory recap (read first, about 10 minutes)

1. Functions: input in, output out

A function is a machine: put in a number x , get out exactly one number $f(x)$. For $f(x) = 2x + 1$, the input 3 gives $f(3) = 2 \times 3 + 1 = 7$. Four functions you will meet again and again:

Straight line	$f(x) = w \cdot x + b$	$f(x) = 3x - 2$ gives $f(2) = 4$
Square (a curve)	$f(x) = x^2$	$f(-3) = 9$ (negative times negative)
ReLU	$f(x) = \max(0, x)$	$\text{ReLU}(-2) = 0$, $\text{ReLU}(3) = 3$
Sigmoid	$f(x) = 1 / (1 + e^{-x})$	from Doc 1: $\text{sigmoid}(0) = 0.5$

2. Graphs and slope

A graph plots each pair $(x, f(x))$ as a point. For a straight line, the **slope** tells how steep it is: how much y changes when x goes up by 1.

Slope of a line	$m = \text{rise} / \text{run} = (y_2 - y_1) / (x_2 - x_1)$	$(1,2)$ and $(3,8)$: $m = 6 / 2 = 3$
Equation of a line	$y = m \cdot x + b$	$m = \text{slope}$, $b = \text{where it crosses the } y\text{-axis}$
Reading the slope	$m > 0$ rises, $m < 0$ falls, $m = 0$ flat	$y = -x + 3$ falls; $y = 4$ is flat

A curve like x^2 has a slope that **changes** from point to point. Measuring that is called a derivative (Doc 10).

3. Sigma notation: a short way to write sums

The Greek letter Sigma means **add up**. The letter i is a counter: start at the bottom value, stop at the top value, and add the expression each time.

Meaning	$\sum_{i=1}^n x_i = x_1 + x_2 + \dots + x_n$	n values added together
Example	$\sum_{i=1}^3 i^2 = 1 + 4 + 9 = 14$	put $i = 1, 2, 3$ into i squared
Mean (average)	$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$	add everything, divide by how many
Constants come out	$\sum c x_i = c \sum x_i$	sum of $2x = 2$ times sum of x

Why ML needs this: a model is a function (features in, prediction out). A weight is a slope. Training follows slopes downhill (Doc 13). The loss adds up errors over many examples with Sigma, and a neuron's input is a weighted sum.

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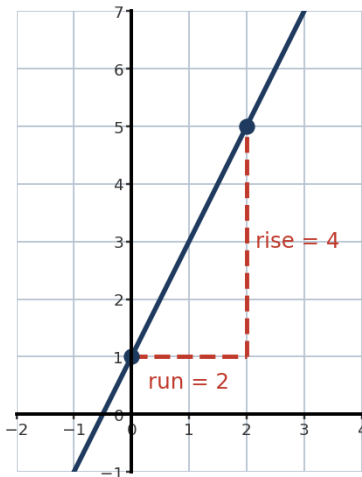
Worked examples: every cell filled in. Read them, then cover the answers and redo them.

Example 1: Function tables

x	-2	-1	0	1	2
$f(x) = 2x + 1$	-3	-1	1	3	5
$h(x) = x^2 - 1$	3	0	-1	0	3

f goes up by 2 every step (constant slope 2). h goes 3, 0, -1, 0, 3: a curve with changing slope.

Example 2: Graph and slope of $y = 2x + 1$



Pick two points on the line: **(0, 1)** and **(2, 5)**.

$$\text{Rise} = 5 - 1 = 4$$

$$\text{Run} = 2 - 0 = 2$$

$$\text{Slope} = \text{rise} / \text{run} = 4 / 2 = 2$$

The line crosses the y-axis at $y = 1$, so **b = 1**, and the equation is **$y = 2x + 1$** : the same function as f in Example 1.

Example 3: Sigma notation

$$\sum_{i=1}^4 i = 1 + 2 + 3 + 4 = 10$$

$$\sum_{i=1}^3 i^2 = 1 + 4 + 9 = 14$$

Mean of a data set: $x = 2, 4, 6, 8$

i	1	2	3	4
x_i	2	4	6	8
Sum of x_i	$2 + 4 + 6 + 8 = 20$			
Mean = sum / 4	$20 / 4 = 5$			

Weighted sum (a neuron's input): weights w, inputs x. Multiply each pair, then add.

i	1	2	3
w_i	1	2	3
x_i	4	5	6
$w_i \times x_i$	4	10	18
Total	$4 + 10 + 18 = 32$		

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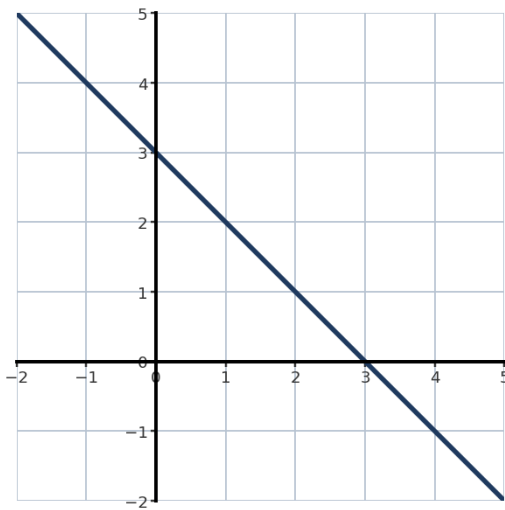
Exercises: use a pencil. Yellow cells are yours to fill in. Name: _____ Date: _____

A. Function tables

x	-2	-1	0	1	2
$f(x) = 3x - 2$					
$g(x) = x^2$					
$\text{ReLU}(x) = \max(0, x)$					

Which of the three functions is a straight line? _____ Its slope is _____

B. Read the slope from the graph



Point 1 (x, y)	
Point 2 (x, y)	
Rise	
Run	
Slope m	
y-intercept b	
Equation	

C. Slope from two points

Two points	Rise ($y_2 - y_1$)	Run ($x_2 - x_1$)	Slope = rise / run
(1, 2) and (3, 8)			
(0, 5) and (4, 1)			
(-2, -3) and (2, 5)			
(1, 4) and (5, 4)			

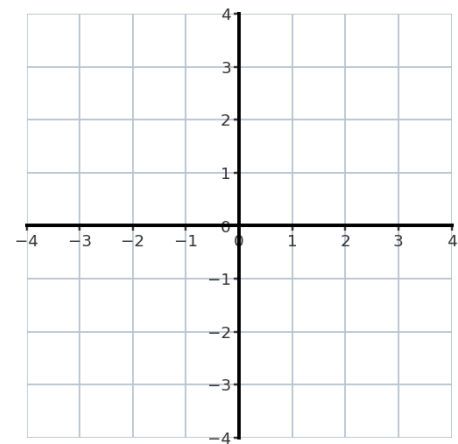
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Exercises, continued

D. Plot two lines and find where they cross

x	-1	0	1	2
$y = 2x - 1$				

x	-1	0	1	2
$y = -x + 2$				



Plot both lines on the graph paper.
They cross at (_____ , _____).

E. Sigma notation

1.	$\sum_{i=1}^5 i$		i	1	2	3	4
			x_i	3	1	4	2
2.	$\sum_{i=1}^4 i^2$		x_i^2				
			Sum of x_i				
3.	$\sum_{i=1}^3 2i$		Mean of x				
			Sum of x_i^2				

F. Weighted sum

i	1	2	3	4
w_i	2	0	1	3
x_i	1	5	4	2
$w_i \times x_i$				
Total				

G. Challenge: how wrong is the line $y = 2x + 1$?

x	1	2	3
Target t	4	4	8
Prediction $p = 2x + 1$			
Error $t - p$			
Squared error			
Sum of squared errors			
Mean = sum / 3			

This average of squared errors is the **mean squared error**, a loss used to train models.

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Answer key: check your work only after finishing

A. Function tables

x	-2	-1	0	1	2
$f(x) = 3x - 2$	-8	-5	-2	1	4
$g(x) = x^2$	4	1	0	1	4
$\text{ReLU}(x)$	0	0	0	1	2

Straight line: $f(x) = 3x - 2$, slope 3.

B and C. Slopes

B: e.g. points (0, 3) and (3, 0): rise -3, run 3, slope -1, $b = 3$, equation $y = -x + 3$.

Two points	Rise ($y_2 - y_1$)	Run ($x_2 - x_1$)	Slope = rise / run
(1, 2) and (3, 8)	6	2	3
(0, 5) and (4, 1)	-4	4	-1
(-2, -3) and (2, 5)	8	4	2
(1, 4) and (5, 4)	0	4	0

D. Plotting

x	-1	0	1	2
$y = 2x - 1$	-3	-1	1	3
$y = -x + 2$	3	2	1	0

The lines cross at (1, 1): check $2(1) - 1 = 1$ and $-1 + 2 = 1$.

E, F and G

E: $1 = 15$, $2 = 1 + 4 + 9 + 16 = 30$, $3 = 2 + 4 + 6 = 12$.

x_i^2	9	1	16	4
Sum x_i	10			
Mean	2.5			
Sum x_i^2	30			

F: products 2, 0, 4, 6, so total = 12.

Prediction p	3	5	7
Error $t - p$	1	-1	1
Squared error	1	1	1
Sum	3		
MSE	$3 / 3 = 1$		

Key takeaways: a function maps input to output; slope = rise / run; $y = mx + b$; Sigma means add up; mean = sum / n; a loss like MSE is a Sigma over squared errors.

Next: Doc 3 - Vectors and matrices from scratch.